Torsion groups of elliptic curves that appear infinitely often over septic, octic and nonic fields

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Torsion groups over a number field of fixed degree

Let E/K be an elliptic curve over a number field. By the Mordell–Weil theorem, the torsion subgroup $E(K)_{\mathrm{tors}}$ is finite, and we write

$$E(K)_{\mathrm{tors}} \simeq \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}, \qquad M \mid N.$$

For simplicity, we denote this group by (M, N).

For a fixed degree d, define

$$\Phi(d) = \{\, E(K)_{\mathrm{tors}} : [K : \mathbb{Q}] = d \,\}.$$

Mazur's theorem determines the case $d = 1$:

$$\Phi(1) = \{(1, n) : n = 1, \ldots, 10, 12\} \cup \{(2, 2n) : 1 \leq n \leq 4\}.$$

We study the subset $\Phi^{\infty}(d) \subseteq \Phi(d)$ consisting of torsion groups which occur for infinitely many $\overline{\mathbb{Q}}$-isomorphism classes of elliptic curves.

What was known?

By Merel's theorem (1996), the set $\Phi(d)$ is finite for every $d \geq 1$.

The full set $\Phi(d)$ is known only in small degrees:

- $d = 1$: Mazur (1977),
- $d = 2$: Kenku–Momose (1988), Kamienny (1992),
- $d = 3$: completed by Derickx–Etropolski–van Hoeij–Morrow–Zureick-Brown (2021),
- $d = 4$: completed by Derickx–Najman (2025),
- $d = 5$: completed by Najman (2026).

We have $\Phi(d) = \Phi^{\infty}(d)$ for $d = 1, 2, 4$, while

$$\Phi(3) = \Phi^{\infty}(3) \cup \{(1, 21)\}, \qquad \Phi(5) = \Phi^{\infty}(5) \cup \{(1, 28), (1, 30), (2, 18)\}.$$

Derickx and Sutherland determined $\Phi^{\infty}(5)$ and $\Phi^{\infty}(6)$ in 2017.

In this work, we determine $\Phi^{\infty}(7)$, $\Phi^{\infty}(8)$ and $\Phi^{\infty}(9)$.

Main theorem

Theorem (Najman–Varivoda)

$$\Phi^{\infty}(7) = \{(1,n) : 1 \leq n \leq 30,\ n \notin \{25,29\}\} \cup \{(2,2n) : 1 \leq n \leq 10\}.$$

$$\begin{aligned}\Phi^{\infty}(8) = {} & \{(1,n) : n \in \{1,\ldots,28,30,32,36\}\} \cup \{(2,2n) : 1 \leq n \leq 12\} \\ & \cup \{(3,3n) : 1 \leq n \leq 4\} \cup \{(4,4n) : 1 \leq n \leq 3\} \cup \{(5,5),(6,6)\}.\end{aligned}$$

$$\Phi^{\infty}(9) = \{(1,n) : n \in \{1,\ldots,28,30,36\}\} \cup \{(2,2n) : 1 \leq n \leq 12\}.$$

The cyclic groups for $d = 7, 8$ were already determined by Derickx–van Hoeij.

Modular curves

Fix $M \mid N$. The modular curve $X_1(M,N)$ is defined over $\mathbb{Q}(\zeta_M)$, and its noncuspidal points parametrize the triples

$$(E, P_M, P_N), \qquad \langle P_M, P_N \rangle \simeq \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z},$$

up to isomorphism. By the Weil pairing, $(M,N) \leq E(K)_{\mathrm{tors}}$ implies $\mathbb{Q}(\zeta_M) \subseteq K$, so $\varphi(M) \mid d$.

The crucial link to modular curves is the following lemma of Derickx–Sutherland.

- If $(M,N) \in \Phi^\infty(d)$, then $X_1(M,N)$ has infinitely many points of degree $d/\varphi(M)$ over $\mathbb{Q}(\zeta_M)$.
- If $X_1(M,N)$ has a function of degree $d/\varphi(M)$ over $\mathbb{Q}(\zeta_M)$, then $(M,N) \in \Phi^\infty(d)$.

So the problem becomes: find low-degree functions, or prove that there are only finitely many low-degree points.

Proof strategy

For each $d = 7, 8, 9$, we do the following steps.

Step 1. First we produce a finite list of candidates $T_d \cup E_d$.
Here T_d is the list of groups appearing in the theorem, and E_d is the excess list that has to be eliminated. This step uses Abramovich's lower bound for the gonality, together with known results about low-degree points on modular curves.

Step 2. For every group $(M, N) \in T_d$, we explicitly construct a modular unit of the required degree $d/\varphi(M)$ on $X_1(M, N)$. This gives a map to $\mathbb{P}^1$, hence infinitely many points of degree $d/\varphi(M)$, as required. The modular unit constructions were carried out using Maarten Derickx's `mdsage` package.

Step 3. For every group $(M, N) \in E_d$, we prove that $X_1(M, N)$ has at most finitely many points of degree $d/\varphi(M)$.

The last step requires most of the work, we used several different methods to tackle it.

Gonality and density degree

Let X/K be a *nice* curve.[1]

- The gonality $\operatorname{gon}_K X$ is the minimum degree of a nonconstant K-rational map $X \to \mathbb{P}^1$.
- The density degree $\delta(X/K)$ is the smallest integer n such that X has infinitely many closed points of degree n over K.

We have

$$\delta(X/K) \leq \operatorname{gon}_K X \leq 2\delta(X/K),$$

with $\delta(X/K) = \operatorname{gon}_K X$ if $\operatorname{rk} J(X)(K) = 0$.

If $(M, N) \in \Phi^{\infty}(d)$, then $\delta(X_1(M,N)/\mathbb{Q}(\zeta_M)) \leq d/\varphi(M)$, so Abramovich's bound gives

$$\frac{325}{2^{15}}[\mathrm{PSL}_2(\mathbb{Z}) : \Gamma_1(M,N)] < \operatorname{gon}_{\mathbb{C}} X_1(M,N) \leq \operatorname{gon}_{\mathbb{Q}(\zeta_M)} X_1(M,N) \leq \frac{2d}{\varphi(M)}.$$

This leaves finitely many pairs (M, N). In the rank-zero cases, we replace $2d/\varphi(M)$ by $d/\varphi(M)$.

[1] A curve is *nice* if it is smooth, projective and geometrically integral.

Constructing modular units

For each $(M, N) \in T_d$, we construct a modular unit of degree $d/\varphi(M)$ on $X_1(M, N)$, defined over $\mathbb{Q}(\zeta_M)$. A modular unit is a function whose divisor is supported only at the cusps.

Starting with a known basis of modular units $f_1, \ldots, f_r$, we take their valuations at cusp places:

$$f_i \longmapsto (v_{C_1}(f_i), \ldots, v_{C_c}(f_i)) \in \mathbb{Z}^c.$$

These vectors generate a lattice. For a product $f = f_1^{e_1} \cdots f_r^{e_r}$,

$$\deg f = \frac{1}{2} \sum_{j=1}^{c} \deg(C_j)\, |v_{C_j}(f)|.$$

Thus finding a modular unit of the required degree becomes a lattice search for an element with required weighted ℓ_1 norm.

The constructions were carried out using Maarten Derickx's `mdsage` package. For every group in T_7, T_8 and T_9, we find a modular unit of the required degree.

Eliminating the excess candidates

For each $(M, N) \in E_d$, we have to show that $X_1(M, N)$ has only finitely many points of the relevant degree. The arguments fall into a few categories.

1. **Finite-field gonality computations.**
   We use the Derickx–Terao `CurveArith` Magma package to prove lower bounds for $\mathrm{gon}_{\mathbb{F}_p} X_1(M, N)$.
2. **Cover searches.** For $X_1(43)$, $X_1(53)$, and $X_1(57)$, we prove $\mathrm{gon}_{\mathbb{F}_2} X_1(N) \geq 19$ by searching Riemann–Roch spaces for carefully chosen covering sets of divisors.
3. **Maps to smaller modular curves.**
   If $X \to Y$ is a finite map and Y has only finitely many points of degree $\leq d$, then the same is true for X. This eliminates several remaining cases.
4. **Two special cases: $X_1(32)$ and $X_1(37)$.** These require separate arguments and are the main exceptional cases in the elimination.

1. Finite-field gonality computations

Computations using `CurveArith` by Maarten Derickx and Kenji Terao.

For $K = \mathbb{Q}(\zeta_M)$, choose $\mathfrak{p} \mid p$ of good reduction and set $k_\mathfrak{p} = \mathcal{O}_K/\mathfrak{p}$. Then $\operatorname{gon}_K X \geq \operatorname{gon}_{k_\mathfrak{p}} X_\mathfrak{p} = \gamma$.

Curve	p	γ	Time (h)	Curve	p	γ	Time (h)
$X_1(2,22)$	3	8	0.009	$X_1(2,32)$	3	≥ 10	0.340
$X_1(2,24)$	7	8	0.170	$X_1(2,34)$	3	≥ 10	0.790
$X_1(2,26)$	3	12	0.384	$X_1(2,36)$	5	≥ 10	1.888
$X_1(2,28)$	3	12	0.325	$X_1(41)$	2	≥ 10	0.364
$X_1(3,15)$	2	6	0.001	$X_1(42)$	5	≥ 10	0.981
$X_1(3,18)$	7	6	0.004	$X_1(44)$	3	≥ 10	0.322
$X_1(4,16)$	5	8	0.006	$X_1(45)$	2	≥ 10	0.336
$X_1(5,10)$	11	3	< 0.001	$X_1(46)$	3	≥ 10	0.831
$X_1(6,12)$	7	6	0.009	$X_1(48)$	5	≥ 10	1.160
$X_1(2,30)$	7	≥ 10	3.902	$X_1(50)$	3	≥ 10	0.932

All these Jacobians have rank zero over K, so $\delta(X/K) = \operatorname{gon}_K X \geq \gamma$.

2. Cover searches over $\mathbb{F}_2$

Suppose $X = X_1(43)$ admits a map $f : X \to \mathbb{P}^1$ of degree at most 18. Since $\#X(\mathbb{F}_2) = 21$ and $\#\mathbb{P}^1(\mathbb{F}_2) = 3$, the pigeonhole principle puts at least 7 rational points in one fiber. Move this fiber to ∞: the pole divisor D has degree at most 18 and contains these 7 points.

We only need to cover effective divisors D of degree at most 18 whose support contains at least 7 rational points. Choose a small set $\mathcal{S}$ so that for every such D, some $A \in \mathcal{S}$ and $a \in (\mathbb{Z}/43\mathbb{Z})^\times$ satisfy

$$\langle a \rangle D \le A \qquad \Longrightarrow \qquad L(\langle a \rangle D) \subseteq L(A).$$

We choose the covering divisors to keep the Riemann–Roch spaces $L(A)$ small, then search them for nonconstant functions of degree at most 18 using 32 parallel workers.

Curve	Genus	$\lvert\mathcal{S}\rvert$	Wall time (h)
$X_1(43)$	57	971	5.53
$X_1(53)$	92	329	55.81
$X_1(57)$	85	308	43.58

No such functions occur. Thus $\mathrm{gon}_{\mathbb{F}_2} X_1(N) \ge 19$ and $\delta(X_1(N)/\mathbb{Q}) \ge 10$ for $N = 43, 53, 57$.

3. Maps to smaller modular curves

If $X \to Y$ is a finite map over $\mathbb{Q}$ and Y has only finitely many points of degree $\leq d$, then the same holds for X.

For $X_1(63)$, take $\Delta = \{\pm 1, \pm 20, \pm 22\} \subseteq (\mathbb{Z}/63\mathbb{Z})^\times$. We use the intermediate curve

$$\underbrace{X_1(63)}_{g=97,\ r_{\mathrm{an}}=2} \longrightarrow \underbrace{Y = X_\Delta(63)}_{g=25,\ r_{\mathrm{an}}=0} \longrightarrow X_0(63).$$

Here r_{an} is the analytic rank of the Jacobian.

- Zywina's `Modular` package constructs a model of Y from modular subgroup generators.
- Derickx–Terao's `CurveArith` computes $\mathrm{gon}_{\mathbb{F}_2} Y_{\mathbb{F}_2} = 12$.

Since $J(Y)(\mathbb{Q})$ has rank zero,

$$\delta(Y/\mathbb{Q}) = \mathrm{gon}_{\mathbb{Q}} Y \geq \mathrm{gon}_{\mathbb{F}_2} Y_{\mathbb{F}_2} = 12.$$

Thus Y, and hence $X_1(63)$, has only finitely many points of degree ≤ 9. This excludes $(1, 63)$ from $\Phi^\infty(9)$.

The curve $X_1(32)$

This curve has a degree-8 modular unit, and hence infinitely many points of degree 8. We want to rule out infinitely many points of degree 9.

Since $J_1(32)(\mathbb{Q})$ has rank zero, it suffices to rule out functions of degree 9 over $\mathbb{Q}$. We use a criterion of Derickx–van Hoeij, which requires finite-field computations together with conditions on reduction and lifting. Our computational input is the following.

Over $\mathbb{F}_3$, we construct $22\,606$ effective divisors A of degree at most 18, such that every $\mathbb{F}_3$-rational effective divisor D of degree 9 satisfies $D \leq A$ for some A in this set.

Searching the Riemann–Roch spaces $L(A)$ over $\mathbb{F}_3$, we find:

- no functions over $\mathbb{F}_3$ of degrees $1, \ldots, 7$ or 9;
- 56 distinct pole divisors of degree-8 functions over $\mathbb{F}_3$ – no pair of which can be dominated by a degree 9 divisor.

The conclusion is

$$(1, 32) \in \Phi^{\infty}(8), \qquad (1, 32) \notin \Phi^{\infty}(9).$$

The curve $X_1(37)$

Here $J_1(37)(\mathbb{Q})$ has rank 1. It is known that

$$\operatorname{gon}_{\mathbb{Q}} X_1(37) = 18,$$

so the general bound only gives

$$\delta(X_1(37)/\mathbb{Q}) \geq \frac{1}{2} \operatorname{gon}_{\mathbb{Q}} X_1(37) = 9.$$

Thus the gonality argument is inconclusive for ruling out infinitely many degree-9 points.

The only positive-rank simple isogeny factor of $J_1(37)$ comes from

$$\pi : X_1(37) \longrightarrow X_0^+(37) \simeq E, \qquad E : \ y^2 + y = x^3 - x.$$

Using Faltings' theorem, we reduce the problem to ruling out a translate of this elliptic factor inside $W_9^0(X_1(37))$.

The curve $X_1(37)$

For a smooth projective curve X/K, write

$$W_d^0(X) = \{[D] \in \mathrm{Pic}^d(X) : \ell(D) \geq 1\}.$$

This is the image of $X^{(d)}$ under the Abel–Jacobi map.

If X has infinitely many closed points of degree d, then either

1. X admits a K-rational map of degree d to $\mathbb{P}^1$, or
2. $W_d^0(X)$ contains a translate of a positive-rank abelian subvariety of $J(X)$.

For $X = X_1(37)$ and $d = 9$, the first possibility is excluded by gonality. For the second, it suffices to consider the elliptic subvariety $\pi^* E$.

If such a translate existed over $\mathbb{Q}$, it would specialize to a translate in $W_9^0(X_{\mathbb{F}_2})$. We rule this out by a Magma computation.

The computation over $\mathbb{F}_2$

The elliptic quotient satisfies $E(\mathbb{F}_2) \simeq \mathbb{Z}/5\mathbb{Z}$. Choose distinct $P, Q \in E(\mathbb{F}_2)$ and set $D_0 = \pi^*(P - Q)$, a degree-zero divisor on $X_1(37)_{\mathbb{F}_2}$. We verify computationally that $C_0 = [D_0]$ has order 5.

We enumerate all effective $\mathbb{F}_2$-rational divisors D of degree 9. For each one, we check that at least one of the spaces

$$L(D + D_0), \quad L(D + 2D_0), \quad L(D + 3D_0), \quad L(D + 4D_0)$$

has dimension zero.

Thus the classes

$$[D],\ [D] + C_0,\ [D] + 2C_0,\ [D] + 3C_0,\ [D] + 4C_0$$

cannot all belong to $W_9^0(X_{\mathbb{F}_2})$. This excludes the translates arising from the positive-rank elliptic factor.

Consequently, $X_1(37)$ has only finitely many degree-9 points over $\mathbb{Q}$, and $(1, 37) \notin \Phi^\infty(9)$. This completes the proof.

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Thank you for your attention.